PRESIDENT'S OFFICE, REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
KOROGWE DISTRICT COUNCIL

MOMBO DIVISION JOINT EXAMINATION SYNDICATE
FORM FOUR BEST TEN STUDENTS' EXAMINATION

031 **PHYSICS**

TIME: 3:00 HOURS **WEDNESDAY 1ST APRIL 2026**

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Instructions:

1. This paper consists of sections A, B and C with total of eleven (11) questions
2. Answer all questions in section A and B and any two (2) questions from section C
3. Non programmable calculator and mathematical table may be used.
4. Where necessary the following constants may be used:
   - *Acceleration due to gravity* $g = 10m/s^2$
   - *Specific heat capacity of water* $= 4200\ Jkg^{-1}K^{-1}$

SECTION A (16 Marks)

Answer all questions in this section

1. For each of the items (i) – (x), choose the correct answer among the given alternatives and write its letter beside the item number.

   i. A student wants to test a piece of metal to determine whether it was a magnet or not. What observation should be taken on the metal?
      A. Attracted steel filings
      B. Repel a metal bar
      C. Attracted a magnet
      D. Repelled a known magnet
      E. Remain stationary

   ii. Work is said to be done when:
      A. Force acts upon a body, but the body does not move
      B. Force acts upon the body, but the body moves perpendicular to the applied force
      C. Force acts upon a body and it moves in a direction of applied force
      D. Force acts upon a body and a body and the body does not move in the direction of applied force
      E. Force acts upon a body and the body moves.

   iii. Which physics concept among the following is used when one is testing the purity of milk?

A. Osmosis B. Capillarity C. Relative density D. Permeability E. Upthrust

iv. As waves pass by a duck floating on a lake, the duck bobs up and down but remains in the same place. Why the duck is not carried along by the wave motion?
   A. The waves are transverse
   B. The waves are longitudinal
   C. The waves carry matter and energy
   D. The waves carry energy but not matter
   E. Waves have no enough energy to carry the duck away

v. Why is Mercury preferred in clinical thermometers as a thermometric of a liquid to water and alcohol?
   A. It is denser than other liquids
   B. It is opaque and does not need coloring
   C. It is more sensitive to temperature
   D. It is active and does not wet the glass
   E. Forms an image behind the mirror

vi. Which physical phenomenon is observed when a tea bag is dipped into a cup of hot water?

A. Steaming B. Diffusion C. Osmosis D. Evaporation E.Boiling

vii. A car moving with a velocity of 40km/h can be stopped by applying brakes in 2m. If the same car is moving with the speed of 80km/h, what is the minimum stopping distance?
   A. 16m B. 12m C.8m D.4m E. 2m

viii. The loudness of a note produced by a vibrating object depends on;
   A. Number of vibrations per second
   B. The overtones present
   C. The quality of sound
   D. The wavelength between nodes
   E. The amplitude of vibrations

ix. Lenz's law can be used to predict the
   A. Magnitude of back electromotive force in a circuit.
   B. Magnitude of induced current in a circuit
   C. Direction of applied electromotive force across a circuit.
   D. Direction of the induced electromotive force or back electromotive force in a circuit.
   E. Magnitude of induced electro motive force or applied electromotive force.

x. Which among the following could be the best reason as to why the sky appears blue while being observed from the earth's surface?
   A. Selective scattering of sunlight

B. Regular refraction of sunlight
C. Irregular refraction of sunlight
D. Diffuse refraction of sunlight
E. Selective scattering of moonlight

2. Match the items from **List A** with its corresponding function from **List B** by writing the correct letter to the answer sheet provided.

List A	List B
i. Vectors that act on the same plane	A. Resultant vector
ii. A single vector that is a net effect of two or more vectors of the same kind.	B. Equilibrant vector
iii. A set of coordinates used to calculate the position and velocity of an object.	C. Relative Velocity
iv. A force that has the same magnitude as the resultant vector but act in opposite direction.	D. Resolution of vector
v. A process of splitting or separating a single vector into its components which are usually perpendicular to each other.	E. Absolute velocity
vi. A movement of an object as seen by an observer on another frame of reference.	F. Relative motion
	G. Frame of reference
	H. Coplanar vectors
	I. Orthogonal vectors

SECTION B (54 Marks)

Answer all questions in this section.

3. (a) brief explain the condition necessary for total internal refection to occur.
   (b) Why does an object appear coloured when light falls on to it? explain how will be appeared a red tie with green dot when looked on the blue light.
4. (a) (i) Explain why cathode ray tubes (CRT) are evacuated.
   (ii) What happens to the CRT (Cathode ray tube) when a gas is maintained?
   (b) How are hard x-rays produced?
5. (a) You observe a stone and leaf place in a river. One sinks rapidly, and the other gently floats on the water's surface. Explain the factors responsible for this difference in behavior. (4 marks)
   (b). The input force of 11 N acting on the effort arm of a lever moves 0.4 m, which lifts a 40 N weight resting on the resistance arm a distance of 0.1 m. What is the efficiency of the machine? (5marks)
6. (a) With the aid of diagram explain how does the size of a gap in a barrier affect the diffraction of waves? (05marks)
   (b) A string of length 1.2m is stretched and made to vibrate so that stationary wave consisting of two loops is produced. Determine the wavelength of the wave (04marks)
7. (a) Use a well labelled diagram to explain the working principle of a hydraulic braking system (05 marks)

(b) A uniform pencil AB weighing 40g can be balanced horizontally on a knife edge at 2cm from the end A when a mass of 60g is hung from this end. what is the length of a pencil? (04 marks)

8. (a) The efficiency of a press is 75%. If the radius of the load piston is 3cm while that of the effort piston is 1.5cm. calculate the mechanical advantage of the press. (4 marks)

   (b) A boy with a mass 50kg jumps out of a rowing boat of mass 300kg on to the bank with horizontal velocity of 3m/s. with what velocity does the boat begin to moves backwards?

SECTION C:(30 marks)

Answer two (2) questions from this section.

9. (a) What is the relationship connecting the frequency of sound source with its wavelength and the speed of sound in air? (3 marks)

   (b) If 5s elapse between a lightning flash and the clap of thunder, how far away is the storm? (5 marks)

   (c) A string is set vibrating by plucking at a point one-quarter of the length from the end.

   (i) Draw a sketched diagram showing nodes and antinodes in a vibrating string. (3 marks)

   (ii) Write the expression of the frequency note produced by the vibrating string.

10. (i) State the laws of electromagnetic induction.

    (ii) Mention two advantages of alternative current (a.c) generators over direct current (d.c) generators.

    (b) A transformer with 4800 turns in the primary coil is designed to work from 240V a.c mains to give a supply at 8V in order to ring a bell.

    (i) What would happen if the transformer were connected to 240V d.c mains?

    (ii) Briefly explain why the primary current increases when a bell is being rung?

11. (a) (i) Name three (3) basic radiations that occur naturally.

    (ii) Define mass number and atomic number as applied in Radioactivity.

    (iii) Thorium disintegrates in the following manner

    $_{90}Th^{232} \longrightarrow {}_{88}Ra^{228} \longrightarrow {}_{89}AC^{228} \longrightarrow {}_{90}Th^{288} \longrightarrow {}_{88}Ra^{224}$

    State the particles emitted at each part of the disintegration.

    (b) A thorium 228 radioactive source placed in a polythene bottle decayed into a radioactive gas thoron. The decay process was recorded by the Geiger-Muller (G-M) tube as follows: -

The in minutes	0	4	8	12	16	20
Count rate	40	30	20	14	10	7

   i. Draw the graph of the count rate against time.

   ii. From the graph, determine the half – life of thoron.